Physics toys from scrap

Little Miss Momentum

Six science toys you can build from things most people throw away!

Why scrap? Because you do not need a fancy lab to discover physics. Old CDs, bottle caps, paper cups and balloons are enough to show motion, force, energy and light in action. Let's build, test, play and wonder why!



In association with the Indian Women Scientists' Association (IWSA)

By Vidika Mirpuri

Hello, young scientist!

Meet Little Miss Momentum

Momentum means how hard it is to stop something that is moving. Every toy in this pamphlet is about things that move, spin, push and pull. Little Miss Momentum will cheer you on, and every page has the same parts: what you need, how to make it, why it works (in English and हिंदी), tips and safety.

Inside this pamphlet

1. **Jumping Dragonfly** — The suction-cup jumper
2. **Simple DC Motor** — A spinning coil that runs on one battery
3. **CD Hovercraft** — Glide on a cushion of air
4. **Cup Centrifuge** — Spin it and watch the balls fly outward
5. **Spectroscope** — Split white light into a rainbow
6. **Simple Telescope** — Bring faraway things close

Golden rules of the maker

- Ask a grown-up to help with blades, glue guns and sanding.
- Test it, fix it and test it again. Failing is part of the fun.
- Change something and ask "what happens now?"
- Never look at the Sun through a lens or a spectroscope.
- Keep small parts away from little brothers and sisters.
- Clean up and save the scrap for next time.

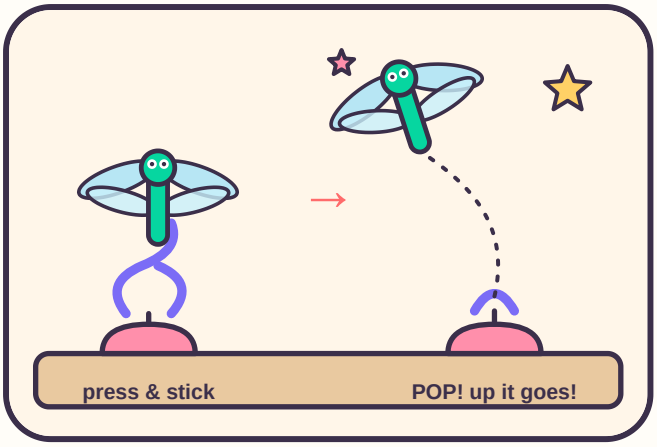
Your scrap shopping list

old CDs · bottle caps · paper cups · balloons · plastic bottles · cardboard tubes · paper clips · beads · old spectacle lenses

1 Jumping Dragonfly

The suction-cup jumper

Difficulty ★☆☆



You will need

- [ ] A rubber suction cup (the kind used for hooks or soap holders)
- [ ] 2 strips of stiff plastic cut from a bottle or folder, about 1 cm × 10 cm
- [ ] Thin card, foam sheet or plastic for the dragonfly body and wings
- [ ] Glue or double-sided tape, scissors, colour pens

Let's make it!

1. Cut a dragonfly body and four wings from the card or foam. Colour it in!
2. Bend each plastic strip into an arch. These are the springy legs.
3. Stick the two arches to the top of the suction cup so they bend like two bent knees.
4. Stick the dragonfly on top of the legs.
5. Wet the rim of the cup. Press the dragonfly down hard on a smooth table until the cup sticks and the legs bend.
6. Let go, step back and wait. After a few seconds the cup lets go and the dragonfly pops up!

Why does it work?

When we press the suction cup, the air under it comes out. The atmospheric pressure outside keeps the cup stuck to the table. At the same time, the plastic legs bend and store elastic potential energy. When the suction cup suddenly comes off, this stored energy is released. It changes into kinetic energy, which makes the animal jump.

जब हम suction cup को दबाते हैं, तो उसके नीचे की हवा बाहर निकल जाती है। बाहर का atmospheric pressure cup को मेज़ से चिपकाकर रखता है। साथ ही, plastic की टाँगें मुड़ती हैं और उनमें elastic potential energy जमा होती है। जब suction cup अचानक छूटता है, तो यह energy बाहर निकलती है। यह kinetic energy में बदल जाती है और dragonfly ऊपर कूदता है।

Atmospheric pressure + Elastic potential energy → Kinetic energy

Fix-it tips

Use a clean, flat, smooth surface like a tile or glass. If the cup will not stick, wash the dust off it. Stiffer plastic legs store more energy, so they jump higher.

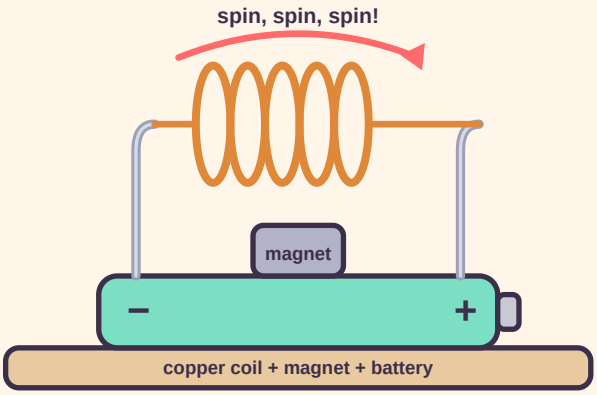
Safety first!

Never aim the jumper at anyone's face. Keep it away from small children.

2 Simple DC Motor

A spinning coil that runs on one battery

Difficulty ★★☆



Why does it work?

The battery sends electric current through the copper coil. The magnets create a magnetic field. When current flows through the coil inside this magnetic field, a force acts on the coil. This force makes the coil move. The half-covered wire works like a simple commutator, helping the coil keep rotating in the same direction.

Battery copper coil में electric current भेजती है। Magnets एक magnetic field बनाते हैं। जब current इस magnetic field में coil से गुजरता है, तो coil पर एक force लगता है। यह force coil को घुमाता है। आधा ढका हुआ wire एक छोटे commutator की तरह काम करता है और coil को लगातार एक ही दिशा में घूमने में मदद करता है।

Electric current + Magnetic field → Force → Rotation

You will need

- [ ] 1 AA battery
- [ ] 1 small, strong round magnet
- [ ] About 1 m of enamelled copper wire (magnet wire)
- [ ] 2 large paper clips
- [ ] Sandpaper, tape and a marker pen to wind the wire around

Let's make it!

1. Wind the wire 8–10 times around the marker to make a coil. Leave about 4 cm of straight wire at each end.
2. Slip the coil off. Wrap the ends around the coil once so it keeps its shape. The two straight ends should stick out opposite each other, like an axle.
3. Sand the enamel off one end completely. On the other end, sand off the enamel on one half only (colour the other half with a marker so you remember!).
4. Straighten the paper clips and bend a small loop in the top of each. Tape one to each end of the battery.
5. Place the magnet on the battery under the coil.
6. Rest the coil's axle in the two loops and give it a gentle flick. It spins!

Fix-it tips

Make the coil balanced, with the loops centred on the axle. Keep the magnet close to the coil. The bare copper must touch the paper clips. If it only buzzes, give it a push to start.

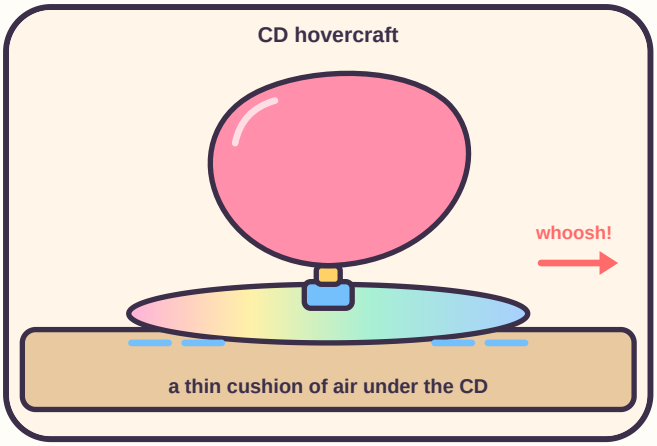
Safety first!

The battery and wire can get warm, so take the battery off when the coil is not spinning. An adult should do the sanding. Keep magnets away from phones, cards and small children.

3 CD Hovercraft

Glide on a cushion of air

Difficulty ★☆☆



You will need

- [ ] 1 old CD or DVD
- [ ] 1 push-pull sports bottle cap
- [ ] 1 balloon
- [ ] Strong glue (hot glue works best) or clay
- [ ] Stickers or pens to decorate

Let's make it!

1. Clean the CD. Keep the shiny side down.
2. Glue the bottle cap over the hole in the middle of the CD. Run glue all the way around so no air can leak. Let it dry fully.
3. Blow up the balloon and pinch the neck so the air stays in.
4. Stretch the neck of the balloon over the base of the cap. Keep the cap closed.
5. Put the hovercraft on a smooth, flat surface such as a table, tile or glass.
6. Pull the cap open and give a gentle push. Whoosh! It glides until the balloon runs out of air.

Why does it work?

The air inside the balloon is under pressure. When the air comes out, it spreads underneath the CD and creates a thin air cushion. This reduces the contact between the CD and the table. As a result, friction becomes very small. When we push the CD, it can move easily because there is very little friction stopping it.

Balloon के अंदर की हवा pressure में होती है। जब हवा बाहर निकलती है, तो वह CD के नीचे फैलकर एक पतली air cushion बनाती है। इससे CD और मेज़ का contact कम हो जाता है। इसलिए friction बहुत कम हो जाता है। जब हम CD को धक्का देते हैं, तो वह आसानी से आगे बढ़ती है क्योंकि उसे रोकने वाला friction बहुत कम होता है।

Air pressure → Air cushion → Less friction

Fix-it tips

If it will not glide, air is leaking, so add more glue around the cap. It works best on smooth tables, tiles or glass, not on carpet.

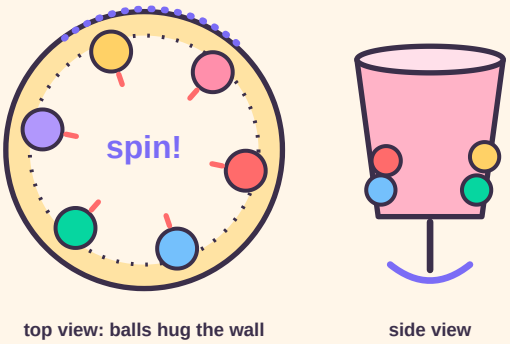
Safety first!

Balloons are a choking risk for small children, so use them with an adult nearby. Throw away burst balloon pieces. Do not chew them.

4 Cup Centrifuge

Spin it and watch the balls fly outward

Difficulty ★☆☆



top view: balls hug the wall

side view

You will need

- [ ] 2 paper or plastic cups of the same size
- [ ] 6–8 small balls or beads (use different colours!)
- [ ] 1 pencil
- [ ] Tape and scissors

Let's make it!

1. Put the beads in one cup.
2. Tape the second cup upside down on top, rim to rim, to make a closed capsule.
3. Poke a small hole in the middle of the bottom cup and push the pencil through. Tape it firmly so it cannot wobble.
4. Hold the pencil between your palms and spin it quickly, like a top.
5. Look at the beads while you spin. They move to the outer wall!
6. Stop and watch them drop back down. Spin slowly, then fast, and compare.

Why does it work?

When we spin the cups, the balls move in a circle. The cup pushes the balls towards the centre. This force is called centripetal force. The balls naturally want to keep moving in a straight line because of inertia. This makes them press against the outer wall of the cup. A real centrifuge uses the same basic idea.

जब हम cups को घुमाते हैं, तो balls गोल रास्ते में घूमती हैं। Cup की दीवार balls को centre की ओर धकेलती है। इस force को centripetal force कहते हैं। Balls inertia के कारण सीधी दिशा में चलना चाहती हैं। इसलिए वे cup की बाहरी दीवार की ओर दबती हैं। असली centrifuge भी इसी principle पर काम करता है।

Centripetal force + Inertia → Circular motion

Fix-it tips

Use equal-sized beads for a clear result. Spin faster to see the beads press harder against the wall. Try different sized beads or a little rice.

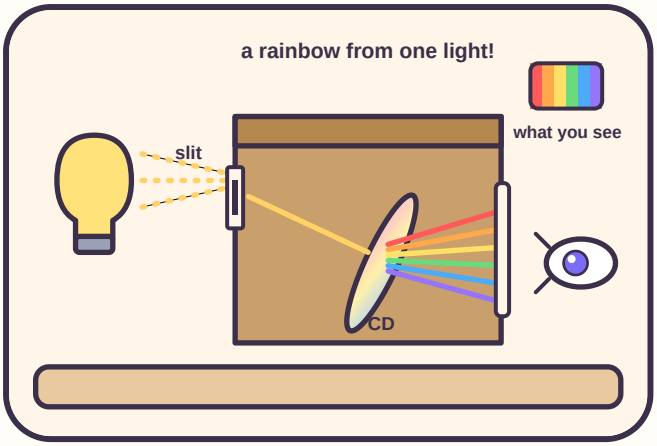
Safety first!

Tape the cups very tightly so beads cannot fly out. Spin it away from faces and keep beads away from small children.

5 Spectroscope

Split white light into a rainbow

Difficulty ★★☆



You will need

- [ ] A cardboard box or tube (a long chip tube or a small box works well)
- [ ] 1 old CD
- [ ] Black paper or black paint
- [ ] A craft knife or blade (adult only) and a ruler
- [ ] Tape and scissors

Why does it work?

White light is made of many different colours. When light enters through the narrow slit and hits the CD, the tiny grooves on the CD act like a diffraction grating. They spread different wavelengths of light in different directions. This separates the white light and produces a spectrum.

सफ़ेद रोशनी कई अलग-अलग रंगों से मिलकर बनी होती है। जब रोशनी पतली slit से अंदर आकर CD पर पड़ती है, तो CD की छोटी-छोटी grooves एक diffraction grating की तरह काम करती हैं। वे अलग-अलग wavelength की रोशनी को अलग-अलग दिशाओं में फैला देती हैं। इससे सफ़ेद रोशनी अलग हो जाती है और spectrum बनता है।

Light + Diffraction grating → Spectrum

Let's make it!

1. Colour or line the inside of the box with black paper so it is dark inside.
2. Cut a very thin slit, about 1 mm wide, at one end. Two strips of black card with a tiny gap between them also work well.
3. Cut a small viewing hole on the side near the slit, about 2 cm wide.
4. Fix the CD inside at the far end at an angle, with the shiny side facing the viewing hole.
5. Point the slit at a lamp. Look through the viewing hole and tilt the box until you see a coloured band.
6. Compare different lamps: an LED, an energy-saving lamp and a torch. Do the colours look the same?

Fix-it tips

A thinner slit gives a sharper rainbow. Work in a dim room. Move the CD angle a little until the band appears.

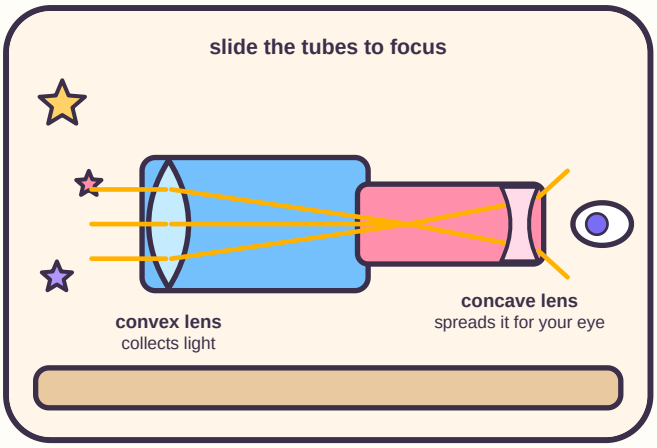
Safety first!

NEVER look at the Sun through the spectroscope. Use lamps or the sky only. An adult should handle the blade.

6 Simple Telescope

Bring faraway things close

Difficulty ★★☆



You will need

- [ ] 1 convex lens (a reading glass or magnifying glass, thicker in the middle)
- [ ] 1 concave lens (a negative spectacle lens, thinner in the middle)
- [ ] 2 cardboard or plastic tubes, one slightly narrower so it slides inside the other
- [ ] Tape, card rings and a marker

Let's make it!

1. Find your lenses. The convex one is thicker in the middle, and the concave one is thinner in the middle.
2. Fix the convex lens at the front of the wide tube using a card ring and tape.
3. Fix the concave lens at the end of the narrow tube the same way.
4. Slide the narrow tube into the wide one.
5. Look through the concave lens at something faraway, like a tree or a building.
6. Slide the tubes in and out until the picture is sharp.

Why does it work?

A simple telescope uses lenses to collect and magnify light from distant objects. The convex lens at the front collects light and brings it together. The concave lens at the back then changes the direction of the light so that our eye can see a magnified image. This is different from GIT because GIT mainly uses mirrors instead of lenses, but both systems use the same basic idea: collect light, focus it, and form an image.

एक साधारण telescope दूर की चीज़ों से आने वाली रोशनी को lenses की मदद से इकट्ठा करके बड़ा दिखाता है। आगे लगा convex lens रोशनी को इकट्ठा करके एक जगह लाता है। पीछे लगा concave lens रोशनी की दिशा बदल देता है, ताकि हमारी आँख को बड़ी तस्वीर दिखाई दे। यह GIT से अलग है, क्योंकि GIT में ज़्यादातर lenses की जगह mirrors इस्तेमाल होते हैं, लेकिन दोनों में बुनियादी सोच एक ही है: रोशनी इकट्ठा करो, focus करो और तस्वीर बनाओ।

Convex lens collects light + Concave lens bends it → Magnified image

What is GIT? The GROWTH-India Telescope (GIT) is a real research telescope in India. It mainly uses mirrors, not lenses, to collect light and hunt for flashes in the sky that appear and fade, called transients.

Fix-it tips

Put the lens with the longer focal length at the front, and keep the lenses about the difference of their focal lengths apart. The picture is the right way up, but you only see a small patch of sky.

Safety first!

NEVER point it at the Sun. It can seriously damage your eyes. Handle the lenses by the edges.

Challenge corner!

Can you figure it out?

1. Why does the dragonfly wait a few seconds before it jumps?
2. What happens to the motor if you sand both ends of the wire completely?
3. Can your hovercraft carry a coin? What about two coins?
4. Do the beads in the centrifuge move faster when you spin faster?
5. Do all lamps make the same rainbow in your spectroscope?
6. What happens to the picture if you swap the two lenses?

Make it your own

Decorate the toys. Race hovercrafts with friends. Invent a new toy from the scrap you find at home. Teach someone else how to make one. The best way to learn physics is to explain it!

Thank you

Little Miss Momentum was created by Vidika Mirpuri, in association with the Indian Women Scientists' Association (IWSA).

